

Q.1) Explain Network Ecosystem

→ The network ecosystem refers to a collection of different components and entities that work together to provide communication services to end users.

- It includes users, service providers, application and infrastructure that enable data transmission over network.
- The main purpose of a networking ecosystem is to deliver efficient, reliable and scalable services such as internet access, cloud computing and communication.

→ Components of Network Ecosystem :

(i) End User

- The end user is the final consumer of network services. Users can access services through devices such as computers, laptops, smartphones, or tablets.

(ii) Network Provider:

- A Network Provider is an organization that offers communication services over a large geographic area. It manages network infrastructure such as routers, switches, and transmission media. Examples include ISPs like Jio or Airtel.

(iii) Application Provider

(iv) Application Service Provider (ASP)

(v) Content Provider:

- Content providers supply digital content such as video, music and data. Examples include Youtube, Netflix & news websites.

Q.2) Explain Global Network Architecture.

- Global network architecture refers to the structure of interconnected networks that provide communication services across the world. It is mainly based on the Internet Protocol (IP) & consists of multiple routers & communication links.
- At the core of this architecture is the IP backbone network, which connects different networks globally.

⇒ Key Components

(i) Core Network (Backbone)

- The backbone network consists of high-performance routers known as core routers. These routers are interconnected using high-speed optical links & handle large volumes of data traffic.

(ii) Core Routers:

- Core routers operate at the center of the network & are responsible for fast data forwarding across the backbone.

(iii) Edge Routers:

- Edge routers are located at the boundary of the network & connect end users or local network to the core network.

(iv) Aggregation Routers:

- These routers combine traffic from multiple networks & forward it to the backbone network.

(v) Optical Links (WDM):

- High-speed communication is achieved using optical fiber links with technologies like wavelength-division multiplexing (WDM).

⇒ Working:

Data packets travel from user devices to edge routers, then to core routers through the backbone network & finally reach the destination network.

Cloud Computing

MODULE-1

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Q.1) Define cloud computing & identify the main features of Cloud computing.

→ Cloud computing is a technology in which computing services like storage, servers, software & networks are provided over the internet.

- Users can use resources whenever needed and pay only for what they use, without owning physical hardware.

— Main Features of Cloud Computing:

- On-demand Service — Resources are available anytime.
- Pay as you use — Pay only for used services.
- Resource Pooling — Resources are shared among many users.
- Scalability — Resource can increase or decrease easily.
- Cost Effective — No need to buy expensive hardware.

— Types of Cloud Services

- ① IaaS (Infrastructure as a Service)
- ② PaaS
- ③ SaaS

Module-2

Q.1) Identify cloud delivery models & explain each ^{with} examples.

- Cloud delivery models define how cloud services are delivered to users over the internet. These models help organizations choose the correct type of cloud service according to their requirements, cost & control level.
- The three main cloud delivery models are:

i) Infrastructure as a Service (IaaS)

- Infrastructure as a Service provides basic computing resources such as virtual servers, storage, networking & OS.
- In this model, the cloud service provider manages the physical infrastructure, while the user manages the operating system, application & data.
- Use Cases:
IaaS is used for website hosting, backup & recovery & enterprise applications. Eg → Amazon EC2, MS Azure

ii) Platform as a Service (PaaS)

- Platform as a Service provides a ready-made platform for application development.
- It includes operating system, runtime environment, databases, & development tools.
- Developers only focus on coding & deployment.
- Examples:
⊗ Google App Engine ⊗ MS Azure ⊗ Heroku
- Use cases;
PaaS is used for developing web & mobile applications.

III) Software as a Service (SaaS) :

- Software as a Service deliver complete software application over the Internet.
- Users access software using a web browser without installing or maintaining it.
- Examples
 - * Gmail
 - * Google Docs
 - * Salesforce

(Q.) What is IoT? Write any 5 applications of IoT.

⇒ Internet of Things (IoT) refers to a network of physical devices such as sensors, machines, vehicles & appliances that are connected to the internet & can collect, send and receive data.

- These devices communicate with each other without human intervention & help in automation & smart decision-making.
- Eg: A smart AC that automatically adjusts temperature based on room conditions.

⇒ Applications of IoT:

(I) Smart Home -

- IoT is used in smart homes to control devices like lights, fans, AC & security systems remotely.

(II) Healthcare -

- IoT devices monitor patient health such as heart rate, blood pressure, & send data to doctor in real-time.

(III) Smart Cities -

- Used for traffic control, waste management, smart lighting & public safety.

(IV) Agriculture -

- IoT helps in smart farming by monitoring soil moisture, weather conditions & crop health.

(V) Industrial Automation -

- Used in industries for machine monitoring & improving efficiency.

Q.3) Discuss REST & its Constraints

- REST (Representational State Transfer) is an architectural style used for designing web services that communicate over HTTP.
- It enables interaction between Client & Server in a simple & Scalable way.
 - REST is widely used in modern networking, especially in SDN Controllers, where communication between applications & controllers is done using REST APIs.
- Key Features of REST:
- Uses HTTP methods such as GET, POST, PUT, DELETE
 - Lightweight communication using JSON/XML
 - Platform independent.

— REST Constraints: —

- (I) Client-Server Architecture:
 - The Client & Server operate independently. The Client sends request & the Server provides response.
- (II) Statelessness:
 - Each request must contain all required information. The Server does not store Client state.
- (III) Cacheable:
 - Responses can be cached to improve performance & reduce server load.
- (IV) Code on Demand:
 - server can send executable code (like JS) to the client.

Q.) Describe the SDN Architecture.

→ SDN architecture is divided into three main layers:

① Application Plane:-

- The application plane is the top layer of SDN architecture & is responsible for providing network services.

→ Functions of Application Plane:

- ① Network Management: Applications manage network operation such as configuration, monitoring & control.
- ② Traffic Engineering: controls the flow of data to optimize network performance.
- ③ Security Management: provides security services such as firewalls, intrusion detection & access control.
- ④ Load Balancing: Distributes network traffic evenly to avoid congestion.
- ⑤ Policy Implementation: Defines rules & policies that the controller follows to manage the network.

② Control Plane:

- This is the middle layer & act as the brain of the network.
- It is implemented using an SDN controller.
- The controller decides how data should flow in the network.
- It receives instruction from the application plane & sends commands to the data plane.

③ Data Plane:

- This is the bottom layer.
- It consists of network devices like switches & routers.
- It is responsible for forwarding data from one device to another.
- It follows the instructions given by the controller.

4) Explain the importance of Cooperation & coordination among SDN controllers.

→ Software Defined Network (SDN) is a modern networking system where the network is controlled by a software called a controller. This controller manages how data flows in the network.

- In small networks one controller is enough. But in large networks, multiple controllers are used. These controllers must cooperate (work together) & coordinate (manage properly) to control the network efficiently.

⇒ ~~Need for~~ Importance of Cooperation & Coordination

(I) Scalability (Handling Large Network)

- Cooperation allows multiple controllers to share the workload, making it possible to manage large & complex network efficiently.

(II) Reliability & Fault Tolerance

- If one controller fails, other controllers can take over its tasks. This ensures continuous network operation without failure.

(III) Improved Performance

- By dividing work among controllers, network operations become faster & more efficient, reducing delays.

(IV) Global Network View

Coordination helps controllers share information, so each controller has a complete view of the network. This leads to better decision-making.

(V) Efficient Resource Utilization

Controllers can coordinate to use network resources properly, avoiding congestion & wastage.

Q.) What are some real-world applications of SDN?

i) Data Centers:

- SDN is used in data centers to manage large numbers of servers & network devices. It helps in efficient traffic management & resource allocation.

ii) Cloud Computing:

- SDN enables dynamic network configuration in cloud environments. It helps in scaling resource & managing virtual networks easily.

iii) Network Security:

- SDN improves security by allowing centralized control of firewalls & access policies.

iv) Traffic Engineering:

- SDN helps in controlling & optimizing network traffic to reduce congestion & improve performance.

v) Telecommunication Networks:

- Telecom companies use SDN to manage large networks, improve service delivery & reduce operational costs.

vi) Internet Service Providers (ISPs)

- ISPs use SDN to efficiently manage bandwidth & provide better quality of service to users.

Q.5) Explain the fundamental Principles of NFV.

→ Network Function Virtualization (NFV) is a technology that virtualizes network services such as routers, firewalls, and load balancers, using software instead of dedicated hardware.

— NFV allows these services to run on standard servers using virtualization technologies.

— Key Principle of NFV:

- (i) Decoupling of Hardware & Software — NFV separates network functions from proprietary hardware & implements them as software.
- (ii) Virtualization — Network functions are executed as Virtual Machines (VMs) on shared hardware resources.
- (iii) Use of Standard Hardware — NFV uses commodity hardware instead of specialized devices, reducing cost.
- (iv) Scalability & Flexibility — Resources can be increased or decreased dynamically based on demand.
- (v) Automation & Orchestration — NFV supports automated deployment & management of network functions.
- (vi) Efficient Resource Utilization — Hardware resources are shared among multiple functions, improving efficiency.

Q.) Describe the Structure & Components of NFV Infrastructure.

- Network Function Virtualization (NFV) is a technology that runs network functions (like firewall, router, etc.) as software instead of hardware.
- The NFV Infrastructure (NFVI) is the foundation of NFV. It provides the hardware & software environment where virtual network functions (VNFs) run.

⇒ Structure of NFVI:

- NFVI consists of physical resources + virtualization layer that together support virtual network functions.
- It acts as a platform that connects hardware with virtual services.

⇒ Components of NFVI:

1) Physical Resources (Hardware)

- These are the actual hardware components:
 - Compute Resources → Servers, CPU, memory
 - Storage Resources → Hard disks, databases
 - Network Resources → Switches, routers, links.
- These provide basic infrastructure for running VNFs.

2) Virtualization Layer

- Also called hypervisor or virtual machine monitor (VMM)
- It creates virtual machines (VMs) from physical hardware
- Allows multiple VNFs to run on the same hardware.
- Eg: VMware, Hyper-V

3) Network Components

Q) What is a Virtual Private Network (VPN)?

→ A Virtual Private Network (VPN) is a technology that creates a secure & private connection over a public network (like the Internet).

- It allows users to send & receive data safely as if they are connected to a private network.

⑥ VPN works by creating an encrypted tunnel between the user & the network. This ensures that data remains secure & cannot be easily accessed by unauthorized users. It is commonly used to protect sensitive data & maintain privacy while using the internet.

→ Key Features of VPN:

- (I) Security (Encryption) — Data is encrypted, making it secure from hackers & unauth. access.
- (II) Privacy — User identity & online activity are hidden.
- (III) Remote Access — Users can access office or private network from anywhere.
- (IV) Secure Communication — Ensures safe data transfer over public networks.

⇒ Types of VPN:

- (I) Remote Access VPN — Allows individual users to connect to a private network remotely.
- (II) Site to Site VPN — Connects two or more networks (e.g. branch office)

Q.) What is Quality of Service (QoS)?

→ Quality of Service (QoS) refers to the ability of a network to provide reliable & better performance by managing & controlling network traffic.

- It ensures that important data is delivered efficiently with minimum delay, loss, & interruption.
- In modern network, different types of data such as video, voice, & text are transmitted simultaneously.
- QoS helps in prioritizing important traffic, so that applications like video calls & online streaming work smoothly.
- For example, during a video call, QoS ensures that voices and video packets are delivered quickly, while less important data like file downloads may be delayed.

→ Key Parameters of QoS:

- (i) Bandwidth — Amount of data that can be transmitted over the network.
- (ii) Latency (Delay) — Time taken for data to travel from source to destination.
- (iii) Packet Loss — Loss of data packets during transmission.

→ Applications of QoS:

- Video Conferencing
- Voice over IP (VoIP)
- Online gaming
- Streaming Services.

Q) What is the difference between QoS & QoE?

→ Quality of Service (QoS) & Quality of Experience (QoE) are both used to measure network performance, but they focus on different aspects.

- QoS focuses on technical performance of the network.
- QoE focuses on user satisfaction & experience.

⇒ QoE:

- Quality of Experience (QoE) refers to the overall satisfaction of the user when using a network service such as video streaming, online gaming, or video calls.
- QoE depends on how smoothly the service works from the user's perspective. Even if the network is technically strong, poor user experience can result in low QoE.
- Example: If a YouTube video keeps buffering, the QoE is poor even if the network has good bandwidth.

QoS = what network says (technical)
QoE = what you feel (experience)

(Q.) What are IP Performance Metrics?

- IP performance metrics are the parameters used to measure the performance & quality of an IP (Internet Protocol) network.
- These metrics help in evaluating how efficiently data is transmitted from source to destination.
 - They are very important in modern networking to ensure good performance for applications like video streaming, voice calls and online gaming.

⇒ Key IP Performance Metrics:

(i) Bandwidth (Throughput)

- Bandwidth refers to the amount of data that can be transmitted over a network in a given time. Higher bandwidth means faster data transfer.

(ii) Latency (Delay)

- Latency is the time taken by a data packet to travel from source to destination. Lower latency is important for real-time applications like video calls.

(iii) Packet Loss

- packet loss occurs when some data packets are lost during transmission. This reduces the quality of communication.